

Winding Tree

Frequently Asked Questions



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What is Winding Tree?

Winding Tree is a decentralized open-source B2B travel marketplace, a platform where buyers and sellers of travel inventory may find trusted

partners they could not find before, connect directly, and start doing business immediately.

Why was Winding Tree created?

In 2016, we came up with a theory that many travel industry problems (like high commission fees, outdated tech, high barriers to entry, and lack of innovation) are caused by high levels of centralization. We published a manifesto, where we described a vision of a decentralized travel marketplace and received overwhelmingly positive responses. We had no other choice but to create that kind of marketplace.

Is Winding Tree an OTA? A GDS?

Neither. Although Winding Tree has similarities with the big market-making OTAs and GDSs; On Winding Tree:

- buyers and sellers may easily find each other (we call this “discovery”), and
- easily verify the other party’s information (we refer to it as “trust”).

What are the differences? It’s an open platform, like the internet, controlled by the community but developed and maintained by the Winding Tree team and its open-source contributors. Therefore, Winding Tree:

- does not impose any contracts on its users;
- does not retain any information (it can’t be hacked);
- is open source,
- is free to use (apart from small Ethereum fees, about \$2–\$4 dollars for changes made to an organization profile).

On Winding Tree, businesses...

- fully control their information;

- decide who they want to do business with;
- may decide to use or not use any part of the platform, any time.

How does Winding Tree work?

On Winding Tree, a business representative creates an account for their company, verify the validity of the information they provided, and then start doing business with other marketplace participants by either exposing their API (in case it's a supplier, like a hotel) or by accessing supplier APIs, if it's an OTA.

Supplier Flow

1. Create a free Winding Tree account
2. Prove the validity of provided information
3. Expose an API through which travel agencies would be able to book supplier inventory

Travel Agent Flow

What is an API?

API stands for Application Programming Interface. It's a language different software systems use to communicate. An API can take specially formatted commands as input and answer accordingly:

- API request: do you have rooms available on August 3, 2020?
- API answer: Yes, there are 3 different types of rooms available on that date.
- API request: do you have a room available starting on August 3, 2020, for 10 nights?
- API answer: no, there are no rooms that match the request.

Question: I am a hotel manager, where do I get an API?

Answer: If you are using a PMS, CRS, a channel manager, etc., chances are that all of those have APIs you could use to sell your inventory on the Winding Tree marketplace. Some of them are ready for Winding Tree, like eRevMax, and some aren't.

Question: I am a PMS/CRS/channel manager representative. What should we do for our clients to be able to sell their inventory on Winding Tree?

Answer: It's incredibly easy to do, and there are a few levels of commitment you may choose from! Here's the [integration guide](#). It should not take more than a few days to complete it. And we're here to help you to make this process as smooth as possible! Let's reinvent travel together! Contact us at partners@windingtree.com

Question: As an OTA representative, how do I know that my business partners are legitimate businesses?

Answer: Winding Tree provides several mechanisms for establishing trust in the ecosystem: claims, credentials, directories, and Lif stake.

Claims

Claims are facts about a business that its representatives may easily prove. For example, connecting a website to the Winding Tree marketplace is fairly easy, but it presents a strong trust signal, as well as connecting your businesses' social network accounts.

Credentials

If claims can be presented and proven independently, credentials are issued by a third party. On Winding Tree, credentials must be cryptographically

signed by another entity on the marketplace. For example, IATA may easily issue unfalsifiable electronic certificates using this approach.

Directories

Directories are manually verified lists of organizations that match certain criteria. For example, all the companies in the Hotels directory are manually verified. In the future, this will be a community-controlled process.

The best way to show others that your company is real is to add it to an appropriate directory. It is one of the best ways of fighting centralization, too!

Lif Stake

Trust is important. Lif Staking is one of the ways to build trust. Our platform is open-source. We don't act as gatekeepers deciding who can join and who can't. In theory, a spammer could join the platform and behave maliciously. Lif Staking is the solution to that. By using our utility token Lif you can send it to a smart contract, think of it as a safety deposit box, that will hold that utility token for your organization allowing you to signal that you have some skin in the game and are here to do business. It is easy to do if you're a legitimate business, but for a spammer, it would be too costly; if they were to create a hundred accounts, they would have to submit a hundred Lif stakes.

The great news is that you may request to withdraw your stake at any time, so the Lif tokens are still yours! There is a 24-hour waiting period for the Lif to be returned (otherwise spammers would be able to game the system).

Question: Winding Tree is decentralized. What does it mean?

Answer: It simply means that there is no entity that controls the marketplace. Centralized control is exactly what creates problems in the travel industry.

Question: Who owns and controls Winding Tree?

Answer: The money that allowed us to create the first version of Winding Tree, was raised from the community who believed in our ideals, therefore, Winding Tree is not owned by anyone, and controlled by the community of its participants.

Question: Winding Tree is an open-source technology. What does it mean? Why should I care?

Answer: We believe that humanity moves forward thanks to open source technologies. The computer, internet, Linux, Firefox, WordPress, Wikipedia, are all examples of modern open source technologies.

Anyone can freely use the technologies our team created, make copies of it, study and experiment with it, improve it, no one can prevent them from doing so...

We are not alone in our belief that open source tech can change the world for the better. So why open source? Because this leaves you in control, of your data, who can access and use it and the overall security of it is secured. Hacks are a modern risk. With our solution, it presents many more degrees of security than modern centralized solutions.

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FAQs

Question: How do you plan on solving the gridlock Travel Management Companies find themselves in when considering leaving GDSs? They need to have access to the entire travel ecosystem, which currently only the GDS provides?

Answer: We suggest that travel buyers implement hybrid models, combining the Winding Tree inventory with the inventory from their legacy sources. As

more content gets onboarded they will be able to shift progressively to this new distribution channel.

Question: What was the reason for using Ethereum, as opposed to creating a new network specifically for travel?

Answer: Ethereum is a public blockchain with best in class support for smart contracts. It also has the most active community of developers, tools, and educational materials. There is no real need to create separate blockchains, as there is no need to create another internet specifically for travel.

Question: Winding Tree is open source. Under which license?

Answer: We use a few different licenses:

- AGPL for smart contracts
- Apache license 2.0 for libraries and code examples
- Creative Commons Attribution 4.0 International license for all texts found on our websites

Question: I thought the Winding Tree marketplace was free to use, but I am required to send Ether to create an account. Why? How much should I pay?

Answer: Winding Tree does not impose any charges on its users, although Winding Tree uses Ethereum, a public blockchain, essentially a distributed database that is hosted on a few thousand independent nodes. The Ethereum database is free to read, but writes and updates require a small amount of Ether, so the request is added to the database by one of the nodes. Again, the fee goes directly to the “miner”, a node that creates or updates records on Ethereum, not to Winding Tree.

Usually creating or updating an account costs anywhere between \$2–4 US dollars. The price is determined by Ethereum miners. Read more about

Ethereum network fees [here](#).

Ethereum network fee is a small price that we pay to avoid centralization problems. The Internet is also free, but in order to access it, you have to own a computer, pay for electricity and internet connection, etc. Ethereum fees are the same type of expense.

And also, as a reminder, this fee is only for the initial onboarding and any changes to your organization profile. There is no per-booking fee. All bookings made and browsing booking inventory is 100% free.

Question: How does Winding Tree make money if there are no fees?

Answer: Winding Tree creates proprietary products on top of the marketplace, in addition to our consulting business.

Question: How will suppliers be added to the marketplace?

Answer: Suppliers are free to join the marketplace, but in order to sell the inventory they must have an API, which individual suppliers may not have. We are developing tools, like the IMS, that allows some suppliers to connect to Winding Tree. But others will either have to integrate with Glider B2B module, which they can do themselves, since it's open-source, or they have to request their software providers (PMS/CRS/GDS, etc.) to support Winding Tree.

We're continuing to seed the marketplace with more companies, both on the supply and demand side.

Question: What will determine the scale of availability of suppliers adding to the marketplace?

Answer: Seeding a two-sided marketplace is not an easy task, so we need your help! If you want a specific supplier or distributor to join the platform,

please invite them! You can direct them to partners@windingtree.com to get started. Only together we can change the status quo!

Question: Can you please explain what is the technical interdependence between Winding Tree and NDC?

Answer: Winding Tree is a marketplace, where buyers and sellers of travel inventory connect directly. That direct connection usually happens via an API. Winding Tree does not impose any specific API standards (we're in no position to do that), therefore suppliers may use any standard they prefer, NDC or otherwise.

Question: Is Winding Tree abstracting the NDC API or would a travel agent still have to integrate with each airline's specific NDC API and version thereof? As an OTA, will I have to, possibly, use different API standards?

Answer: Lack of standardization in travel is a problem. The only way to solve this is to work on these standards openly, in an open source way, not behind closed doors. That's why we open-sourced our codebase for normalizing different hotel and airline APIs. Let's solve this together!

On Winding Tree, travel agents already have a choice to do one of the following:

Connect to suppliers directly (the hardest)

You will have to integrate different supplier and aggregator APIs into your software.

Normalize content via Glider Aggregator (the best)

Since Glider Aggregator is open-source, you may use it in your software and get normalization for a variety of APIs out of the box ([see our open API standard](#)). We (and others!) will be adding more APIs to it over time. Our goal

is to make Glider Aggregator Open API specification the ultimate normalization engine.

Get content from Glider Aggregator hosted by Winding Tree

We have a hosted version of Glider Aggregator. That anyone is welcome to use free of charge.

How is airfare shopping performed?

Winding Tree is not an OTA, we do not impose any limitation on how airlines sell their inventory. But we created the world's first open-source OTA, Glider, where we show how shopping is performed using ATPCO NDC Exchange. Please see how it works on [Glider](#) and explore Glider source code which will be available on Github.

Question: Will you open this up to car rentals eventually? If so, when?

Answer: Any company can join the platform. [Create an account!](#)

Question: What benefits does Winding Tree offer to an airline or agent compared to an NDC aggregator?

Answer: Winding Tree offers a lot more potential content since it's an aggregator of aggregators. You will still have to connect to suppliers and aggregators, but you will be able to do it in a unified and streamlined way. Also, it is free and monopoly-proof!

Question: Can you explain more about the ability of the platform to handle high volume transactions? It is said that Ethereum is unable to handle more than a few transactions per second. Does every booking require an Ethereum transaction?

Answer: The only time an Ethereum transaction is required is when a travel company creates or updates their account on the Winding Tree marketplace. So one needs just a few transactions every year to keep their business information up-to-date. In other words, Winding Tree is designed in a way that does not require a high volume of Ethereum transactions. Nonetheless, the Ethereum community works hard on improving network scalability, for example, zkSync can already scale to thousands of transactions per second.

Question: How is the Winding Tree Marketplace database accessed?

Answer: At the core of the Winding Tree Marketplace, there is an Ethereum smart contract, a distributed database of all participants. Anyone can access the database in read-only mode. Only account owners (and no one else, not even us) may create or update their information.

Question: How do we add our inventory on Winding Tree?

1. Create an account on <https://marketplace.windingtree.com>
2. Prove that you're a real business
3. Expose your API
4. Profit!

Question: If a small country with historically very high visitor numbers were to integrate this technology at Government level for their airline and their hotels would WT combine their resources to help make that happen?

Answer: Absolutely! Contact us at partners@windingtree.com

Question: This sounds incredibly exciting but how are you going to attract and onboard tourism businesses especially small to medium providers who account for the exciting growth within tourism?

By having low barriers to entry: it's free and easy to use!

Answer: It does not look like you integrate with my preferred Channel Manager. Can that be added?

Should you want your preferred technology provider to integrate into the Winding Tree Marketplace, please do ask them and refer them to our [integration guide](#)! If you are interested in sponsoring the development, we can also do the integration with new partners.

Question: Can a hotel onboard itself to the Winding Tree Marketplace use its own PMS?

Answer: Yes. You would need to expose the API to your service publicly. Here's a [technical guide](#) on how you can do that. If you have questions please feel free to reach out to us!

Question: How do you expect mass adoption to happen, particularly on the hotel/travel supply side? In my experience, suppliers are reluctant to implement new tech because of the burden it can put on operations, or quite simply they don't understand it.

Answer: There are ways to ease the migration, for example in the case of hotels asking channel managers to integrate with the platform. For example, we have an integration with eRevmax channel manager on "Glider Aggregator", therefore any hotel using eRevmax can be connected on-demand. Please get in touch with your eRevmax point of contact for instructions.

Question: What is the incentive for a buyer to use Winding Tree over a traditional OTA? Especially when Expedia, Booking, etc are so well known and trusted?

Answer: Winding Tree is not an OTA, and even having built Glider Travel, we did it just as an example of how easily and quickly anyone can build an OTA on top of Winding Tree.

OTAs and TMCs find trusted suppliers on Winding Tree and source content directly from them. Since there is no middleman, that content is cheaper, and the data flows are not restricted by a chain of intermediaries.

So, travelers find cheaper hotels and flights on OTAs that use Winding Tree, while suppliers prefer working with these OTAs since it improves their customer experience.

Question: I run a hotel booking engine company. We would like to understand how we could use your tech. Is it possible to build our own OTA with all of our properties all under our own brand?

Answers: Absolutely! You can even use our Glider Travel source code and create your own OTA in just a few days! Brand it your way, do whatever you want, it's free! If you need help, feel free to contact us anytime at partners@windingtree.com!

Question: Is the platform accredited by IATA, ARC, etc.?

Answer: Winding Tree is not an OTA or TMC, it doesn't even need a legal entity to operate. Hence no such accreditation is required.

Glider Travel, the open-source OTA example, is operated by a separate company (which is required by law), and is IATA-accredited: #63320235.

Question: Can other vendors such as insurance companies, travel associations, local tour operators, chambers of commerce join?

Yes, the platform is open to all travel (and not just travel!) companies!

Answer: There are complex products and destinations where these actually add value for the foreseeable future for intermediaries. Is it possible to supply the Winding Tree ‘channel’ with such a product as an intermediary, or are you opposed to this?

Any company, travel or not, can join Winding Tree. Anyone is welcome! We have no way of controlling who joins the marketplace, which would make us an intermediary. What Winding Tree does is creates a level playing field, where intermediaries will have to compete with each other, ultimately leaving only those companies that add value, and eliminating rent-seeking middlemen.

Question: The airline I work for is working on legacy systems. If I try to talk them into open-source code and blockchain technology they might get nervous. Can the Winding Tree Marketplace be connected to in a more traditional way?

Answer: Absolutely! We can integrate your airline content into Glider Aggregator. Let’s schedule a call and discuss this!

Question: Will tours and experiences be included for distribution as well?

Any company can already join the marketplace! We’re constantly improving our user interfaces to make all segments of travel companies to be much easily discovered.

Answer: Airlines are beginning to move to no middlemen “sales” entities to direct systems, like NDC platforms, who can connect directly with their systems with any other startup. How can Winding Tree position itself against this?

These new NDC platforms are already becoming the new gatekeepers. While they definitely add value to the travel ecosystem, it’s important that they

should be a part of an open marketplace, to avoid another wave of centralization.

Additionally, on Winding Tree, airlines can reach new travel buyers with near-zero effort in terms of onboarding costs, such as API management, settlement, etc. For travel buyers, it is easier to create direct connections to airlines with the Winding Tree marketplace. If you doubt how easy it is, [see our workshop video](#) where we show the full flow of onboarding from scratch and creating a decentralized airline booking in under one hour.

Still have a question? Reach out to us at partners@windingtree.com. We are here to help.

Sign up to our newsletter to stay up-to-date with the latest. We have many exciting announcements coming in the next few months!

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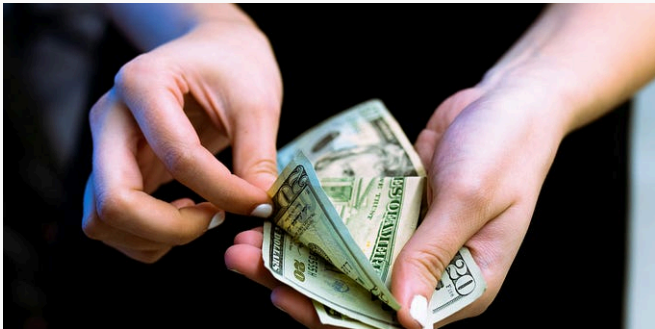
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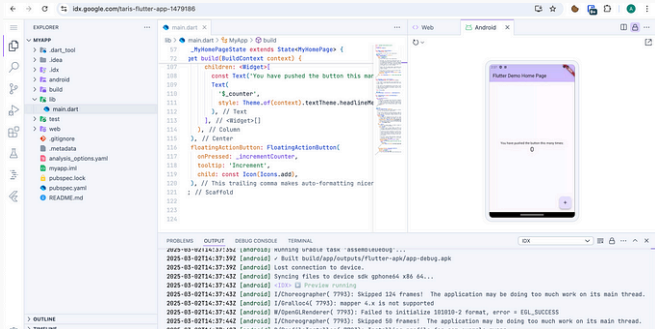
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